

COFFEE SALES ANALYSIS REPORT

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Programme: Cirvee Academy (Project)

Dataset: 2,623 transactions

Tools: Excel and PowerBI

EXECUTIVE SUMMARY

\$83.65K Total Revenue	\$1.31K Average Transaction Value	2623 Total Number of Customer	\$31.89 Average Transaction Value
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This report analyses **2,623 coffee sales transactions** between January and December 2024 using Excel Power Query and Power BI. The shop generated **\$83.65k** in total revenue with an average transaction value of **\$31.89**. Key findings show that Latte and Americano with Milk are top performing products, card payments dominate at **96.61%** and sales peak on Tuesday mornings and in October. Recommendations focus on optimising peak operations, boosting underperforming periods and sustaining revenue growth through data-driven decisions.

PROBLEM STATEMENT

Coffee retail businesses often struggle to identify which products, time periods and payment preferences drive the most revenue. Without data-driven insights, business decisions are made based on assumption rather than evidence. This analysis examines coffee sales transactional data from January to December 2024 comprising 2,623 transactions to uncover patterns in customer behaviour, product performance, payment preferences and temporal trends. The goal is to provide actionable insights that enable coffee retail businesses to optimise operations, boost underperforming periods and sustain revenue growth.

DATA OVERVIEW

The Coffee Sales dataset is a single CSV file sourced from Kaggle, containing transactional records of a retail coffee shop. The dataset comprises 2,623 rows and 6 variables including Date, Cash Type, Card, Money, Coffee Name and Time. Each row represents a unique customer transaction. During the data cleaning and transformation stage, 3 additional columns were engineered, Day Name, Time Range and Day Number to support temporal analysis and uncover sales patterns across different periods of the day and week.

Data Dictionary

Variable	Type	Description	Null
Date	Date	Date of the transaction	0%
Cash type	Varchar	Method of transaction	0%

Card	Varchar	Details of Card Number	3%
Money	INT	Price of Coffee	0%
Time	Datetime	Time of transaction	0%

Data Quality Issues & Decisions

Issue	Volume	Decision
Missing Value	89 (3%)	Replace with 'Not Available'
Inconsistent format	All Text Column	Standardize all text value

METHODOLOGY

A five-stage pipeline using Excel and Power BI was adopted to ensures reproducibility and separation of raw from analytical data.

Stage	Tool	Output	Key Decision
Ingestion	Excel	2623	Load CSV file into Excel Power Query
Exploration	Excel Power Query	Null count and date range audit	3% null was found
Cleaning & Transformation	Excel Power Query	Coffee clean (2623 rows) + 3 new columns added	Standardize text value and replace nulls. Extracted Day Name, Time Range (Morning, Afternoon & Evening) and Day Number from Date Column to support trend analysis
Analysis	Power BI	5 Descriptive questions answered	Connected clean Excel file to Power BI for analysis
Visualization	Power BI	Dashboard	Display insights through charts and graphs

KEY FINDINGS

1. Sales Performance

The coffee shop recorded 2,623 transactions between January and December 2024, generating **\$83.6k** in total sales with an average spend of **\$31.89** per customer. Americano with Milk (621) and Latte (618) were the top performing coffees, indicating strong customer preference for milk-based drinks. Morning sessions drove the highest sales at **\$27.6k**, with Tuesdays being the peak day and Wednesdays the lowest.

Implication:

The shop should prioritize morning operations and milk-based stock, while introducing Wednesday promotions to boost the weakest sales day.

2. Payment Method

Card payments dominated transactions, accounting for 96.61% of all sales compared to just 3.39% via cash. This preference for card payments was consistent across all days and coffee types.

Implication:

The shop should ensure card payment systems are always functional and reliable, as any downtime could significantly disrupt sales. While cash should still be accepted, resources and infrastructure should prioritize seamless digital payment experience.

3. Product Insight

Latte was the top revenue-generating coffee with \$22k from 618 units, followed closely by Americano with Milk at \$19k from 621 units. Espresso was the least popular product, generating only \$2.04k in sales. Payment method showed no significant variation across coffee types.

Implication:

The shop should promote and prioritize Latte and Americano with Milk as flagship products while considering whether to rebrand, discount or replace Espresso to improve its performance. Since payment preference does not vary by coffee type, pricing strategies can be applied uniformly across all products.

4. Temporal Pattern

Tuesday recorded the highest sales across the year while Wednesday consistently underperformed. Morning sessions dominated all time slots, reinforcing the shop's strength during early hours. Sales showed a clear seasonal trend, rising from March, peaking in October and dipping in December.

Implication:

The shop should leverage Tuesday morning traffic with loyalty rewards while running Wednesday promotions to lift the weakest day. The December dip signals a need for festive campaigns or seasonal offers to maintain revenue during the slow period, while October's peak should be prepared for with adequate staffing and stock levels.

5. Business Optimization

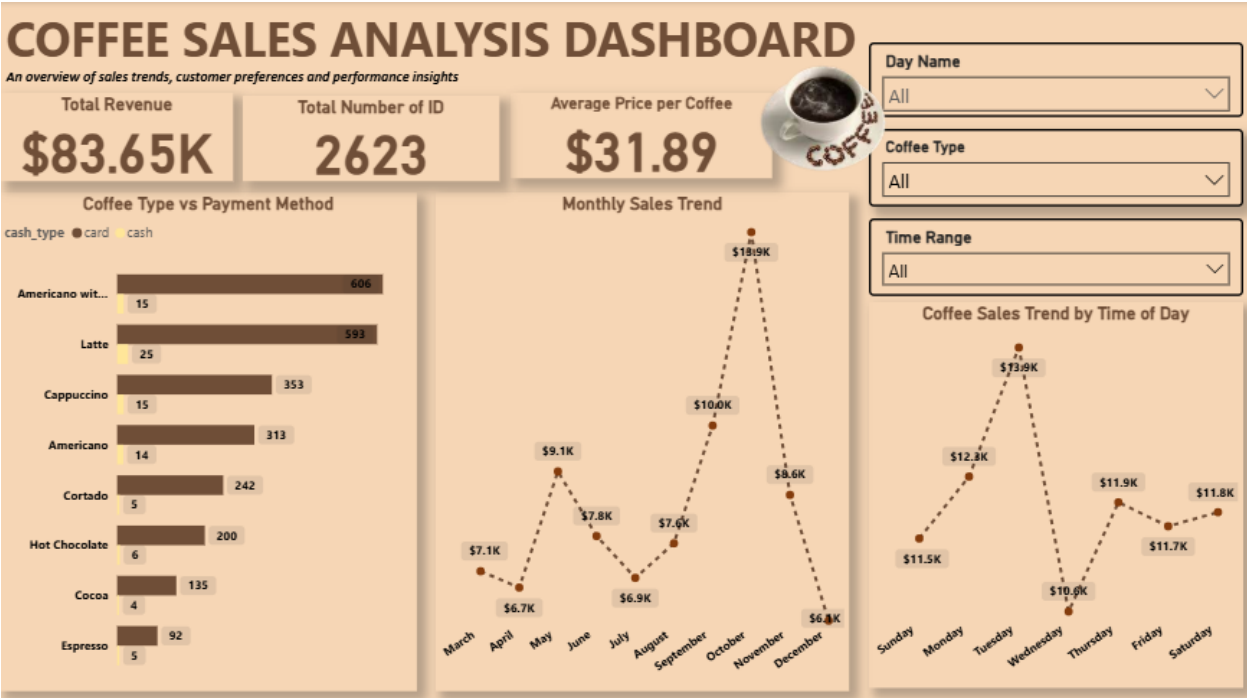
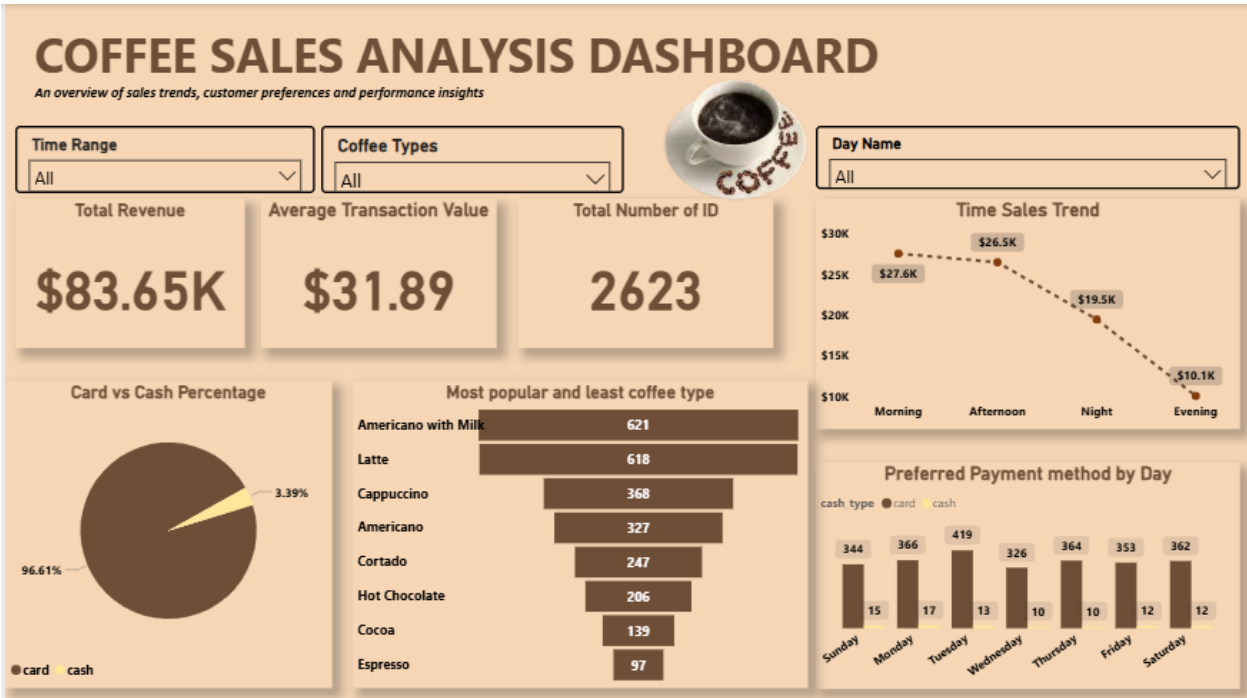
Low-selling products such as **Espresso** and **Cocoa** present opportunities for targeted promotions to increase their contribution to overall revenue. Similarly, **evening time slots** consistently underperformed compared to other periods, representing an untapped opportunity for growth.

Implication: The shop should consider **bundle deals or discounts** on Espresso and Cocoa to stimulate demand. Evening-specific campaigns such as **happy hour promotions or loyalty rewards** could attract more customers during slow periods, ultimately improving overall daily revenue and making operations more efficient across all time slots.

Visual Insight & Dashboard

An interactive Power BI dashboard was published comprising **2 pages** with dynamic slicers for Time Range, Coffee Type and Day Name.

Page	Question Addressed	Visuals
General Sales & Payment Method	What are the overall sales metrics and how do customers pay?	3 KPI Cards, Donut Chart, Bar Chart, Line Chart
Product Insight & Temporal	Which coffee performs best and when do sales peak?	KPI Card, Bar Chart, Line Chart, Area Chart



RECOMMENDATION

Based on the analysis of 2,623 transactions recorded between January and December 2024, the following recommendations are proposed for coffee retail businesses seeking to optimise sales performance and customer engagement:

1. Align Operations with Peak Demand

Businesses should identify their highest performing days and time slots and allocate resources accordingly. In this case, Tuesday mornings drove the most sales, highlighting the importance of adequate staffing and stock during peak periods.

2. Address Underperforming Periods Strategically

Low performing days and time slots such as Wednesdays and evenings should not be ignored. Businesses can introduce targeted promotions, loyalty rewards or happy hour campaigns to stimulate demand during slow periods.

3. Leverage Best Selling Products

Top performing products should be consistently available and well promoted. Businesses should also consider rebranding or bundling low selling products rather than removing them, as targeted promotions can revive interest and increase revenue contribution.

4. Invest in Reliable Digital Payment Systems

With card payments accounting for 96.61% of transactions, businesses must prioritise seamless and reliable digital payment infrastructure as any disruption could significantly impact revenue.

5. Plan Around Seasonal Trends

Businesses should use historical sales data to anticipate seasonal peaks and dips, ensuring adequate preparation during high demand periods and deploying promotional strategies during slow months to maintain steady revenue flow.

CONCLUSION

This report analysed 2,623 coffee sales transactions between January and December 2024, generating a total revenue of \$83.65k. Key findings revealed that Tuesday mornings drive peak sales, Latte and Americano with Milk are the top performing products, and card payments dominate at 96.61%. Sales peak in October and dip in December.

Using Excel Power Query and Power BI, the analysis provides actionable insights to help coffee retail businesses optimise operations, address underperforming periods and sustain revenue growth.